THE PRIVATE CYBER LAB HANDBOOK

LAB 01

Active Directory Environment Baseline, Health Check & Documentation

Windows Server 2019 • AD DS • DNS • RRAS/NAT

Cover field	Complete before submission
Student Name	_____
Roll Number / ID	_____
Course Title	Windows Server & Active Directory Security Lab
Instructor / Institution	_____
Submission Date	_____
Lab Environment	Private VMware Workstation virtual lab

Practical Lab Report • Version 1.0 • Based on observed session results

Document Control

Field	Details
Title	Lab 01 — Active Directory Environment Baseline, Health Check & Documentation
Filename	Lab01_Complete_Documentation.docx
Domain	zroots.local
Domain controller	DC01 — LAN 10.1.1.10
Client	WIN10-01 — 10.1.1.20
Status	Core AD/DNS, client connectivity, external IP and external DNS tests passed.
Evidence	Screenshots are represented by placeholders; insert the original screenshots before submission.

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- 8. Domain Discovery and LDAP SRV
- 9. DNS Registration and Record Remediation
- 10. WIN10-01 Configuration

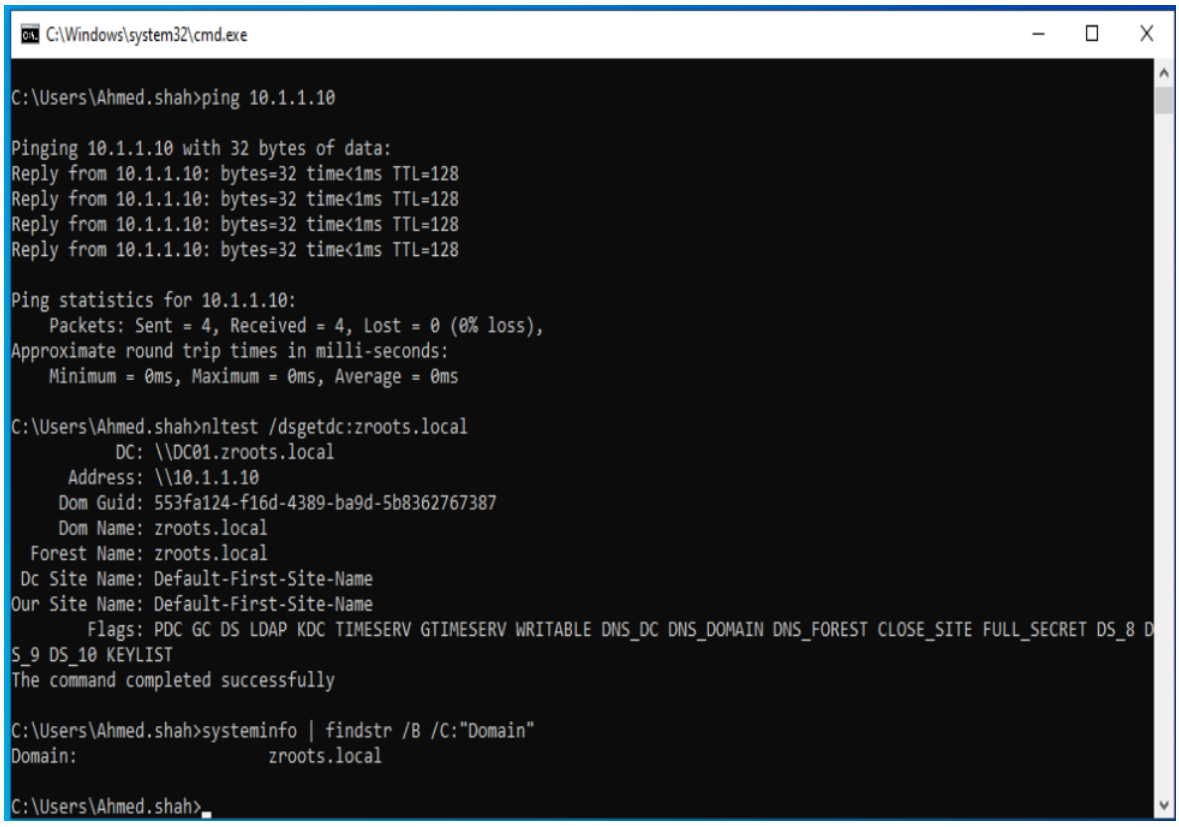


Figure 11.1 — Successful ping to DC01 and domain-controller discovery

- 11. Client Connectivity and Domain Membership

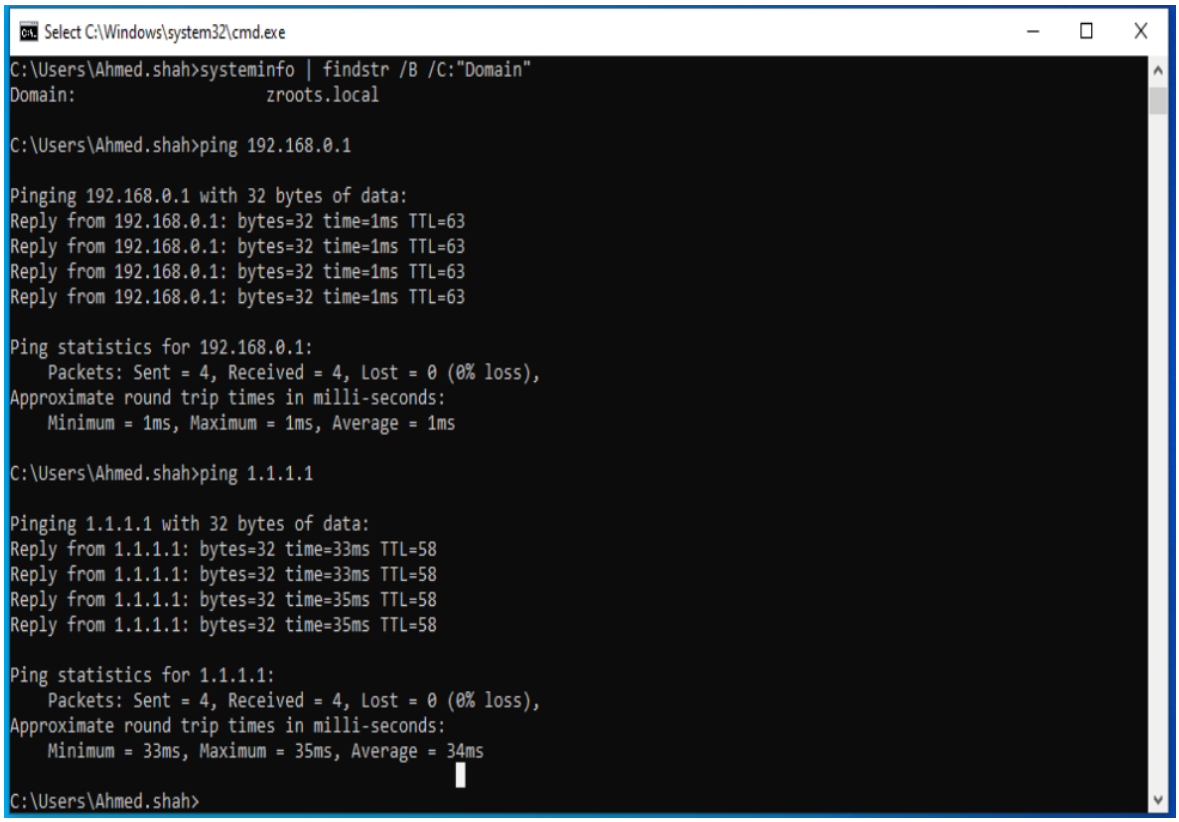


Figure 12.1 — Successful connectivity to the physical router and external IP

Additional observed result: The separately supplied nslookup www.microsoft.com output resolved successfully through DC01 (10.1.1.10), returning IPv4 23.41.38.100 and IPv6 addresses. No separate screenshot of this lookup was provided.

- 12. Internet Connectivity and External DNS
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- 17. Conclusion
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1. Executive Summary

This lab established a baseline for a private VMware lab centered on Windows Server 2019 domain controller DC01 and the Active Directory domain zroots.local. DC01 provides AD DS and DNS and routes traffic between the internal host-only network and a bridged WAN using RRAS/NAT. The session checked services, selected DCDIAG tests, domain-controller discovery, LDAP SRV records, DNS A records and DNS listener settings. WIN10-01 was then tested for correct static addressing, domain membership, reachability to DC01, external IP connectivity and external DNS resolution.

Overall outcome
The observed tests support functional core AD/DNS operation and internet access from WIN10-01 through

the configured routing path. This is a baseline, not a complete security audit; not every endpoint or protocol was tested.

2. Objectives and Learning Outcomes

- Confirm DC01 identity, domain, interface configuration and core service status.
- Understand AD DS, DNS, LDAP SRV records and domain-controller discovery.
- Identify incorrect WAN-address A records and correct them without deleting valid LAN or SRV records.
- Disable WAN interface DNS registration and configure DNS listening to the internal LAN address.
- Verify WIN10-01 static IP, gateway, DNS, domain membership and connectivity.
- Test routed external IP connectivity and external DNS name resolution.
- Record commands, observed results, troubleshooting and security lessons for the cumulative handbook.

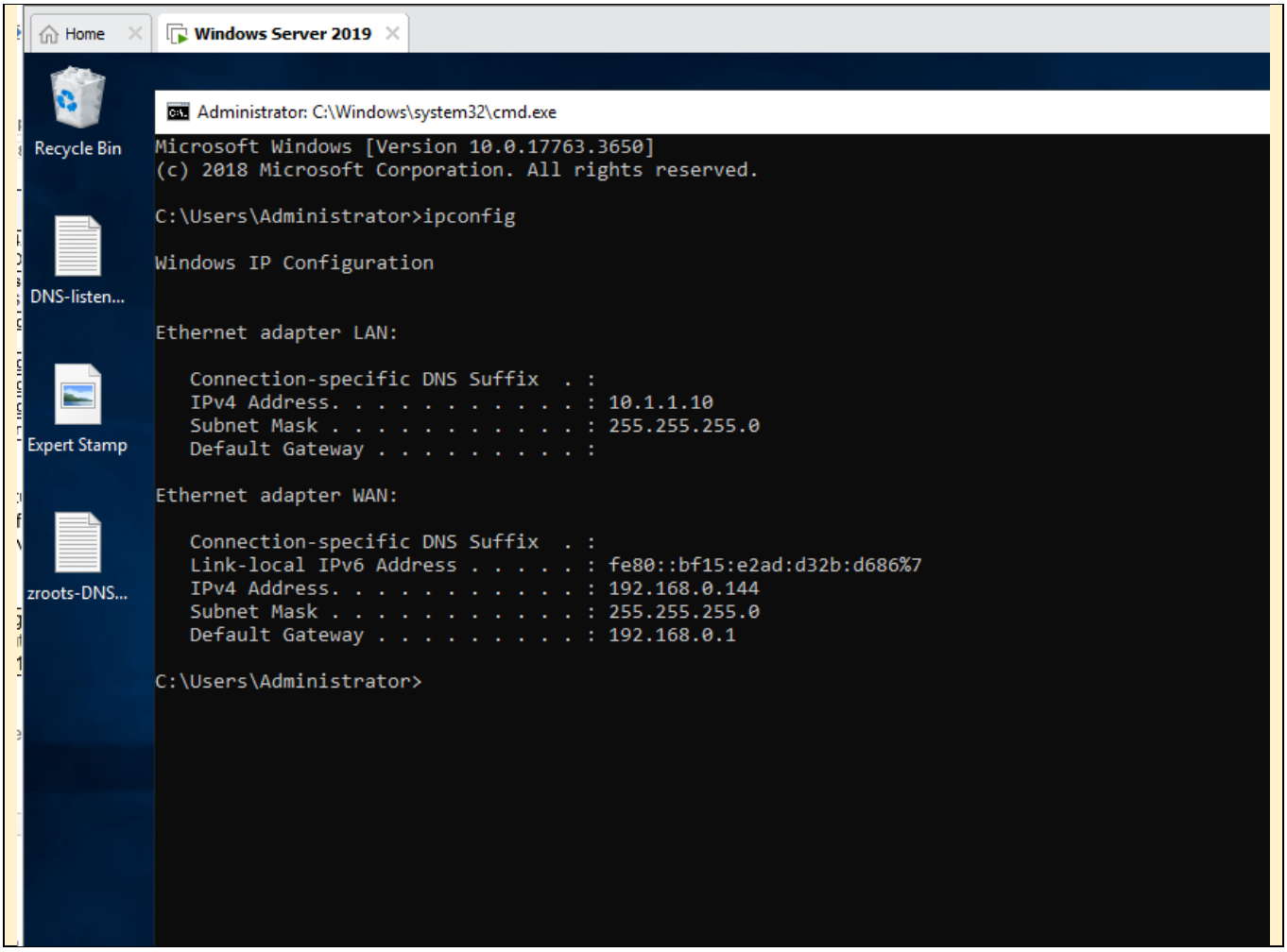
3. Scope, Safety and Rules of Engagement

- This is an authorized private VMware lab. Keep vulnerable targets such as Metasploitable isolated from untrusted networks.
- Do not disable Windows Defender Firewall as a troubleshooting shortcut.
- Do not delete AD SRV records or broad DNS record sets without verifying purpose.
- Capture before-state evidence before DNS changes; this session exported an A-record inventory.
- Avoid unnecessary restarts or disruptive changes; none were needed for the documented steps.
- Use precise filters and verify the result after each change.

4. Lab Topology and Addressing

Component	Network/configuration	Role
Physical router	192.168.0.1	WAN gateway
VMware VMnet0	Bridged	Connects DC01 WAN NIC to physical LAN
DC01 WAN	192.168.0.144/24; gateway/DNS 192.168.0.1; DHCP enabled	Uplink for RRAS
DC01 LAN	MyLabNet host-only; 10.1.1.10/24; no gateway; DNS 127.0.0.1	AD DS/DNS and internal gateway
WIN10-01	10.1.1.20/24; gateway/DNS 10.1.1.10; static	Domain client
Kali Linux	10.1.1.30/24; gateway/DNS 10.1.1.10	Security-testing client
BlackArch Linux	10.1.1.40/24; gateway/DNS 10.1.1.10	Security-testing client
Windows 11	10.1.1.50/24; gateway/DNS 10.1.1.10	Domain client
Metasploitable 2	10.1.1.60/24 in planned topology	Vulnerable target; isolate

VMware Bridged networking attaches the WAN virtual NIC to the physical LAN. RRAS/NAT is a separate Windows Server function translating/routing traffic from the private 10.1.1.0/24 network through the WAN. VMware NAT is not the WAN mode in this design. VMware DHCP should remain disabled on MyLabNet while static addressing is used.



5. Technical Background and Glossary

Term	Definition and relevance
AD DS	Active Directory Domain Services; directory, authentication and identity management.
Domain controller	Server running AD DS; DC01 authenticates clients and provides directory services.
Domain / forest	AD domain and forest naming/administrative structure; this lab uses zroots.local.
DNS	Maps names to addresses and publishes AD service-location records.
A record	DNS hostname-to-IPv4 mapping.
SRV record	DNS record that advertises service target, port, priority and weight.
LDAP	Lightweight Directory Access Protocol; AD directory access commonly uses TCP 389.
NLTEST	Windows command-line tool for domain-controller discovery and related diagnostics.
DCDIAG	Domain Controller Diagnostic utility for AD/DC health tests.
RRAS	Routing and Remote Access Service; Windows Server routing functionality.
NAT	Network Address Translation; translates private-source traffic for an upstream network.
Bridged networking	VM virtual NIC attached to the physical network.
Host-only network	Private virtual network; internet access requires routing

	through a gateway.
Default gateway	Next hop used to reach networks outside the local subnet.
DNS suffix search list	Suffix Windows appends to unqualified names when resolving them.
Dynamic DNS registration	Registration of interface addresses in DNS.
Authoritative/non-authoritative answer	Answer from the zone authority versus an answer returned through recursive/cached resolution.
TTL	Time to Live; how long DNS answers may be cached.
IPv4/IPv6	IP address protocol versions; Microsoft lookup returned both.
ICMP/ping	Echo request/reply used for basic reachability; may be filtered.
DHCP/static IP	Automatic network lease versus manually configured network settings.
NetBIOS over TCP/IP	Legacy Windows networking/name-service support, shown enabled on client.
Diagnostic test record	A record such as dcdiag-test-record may be created during diagnostics; verify before removal.

6. Prerequisites

- DC01 and WIN10-01 available in VMware Workstation.
- DC01 is Windows Server 2019, domain controller for zroots.local.
- MyLabNet is 10.1.1.0/24; DC01 LAN is 10.1.1.10.
- DC01 WAN is bridged and uses 192.168.0.144 with router 192.168.0.1.
- RRAS is running; WAN is Public with NAT enabled; LAN is Private.
- WIN10-01 should use 10.1.1.20/24, gateway 10.1.1.10 and DNS 10.1.1.10.

7. DC01 Baseline and Service Health

7.1 Server identity and addressing

```
hostname
```

Purpose: Displays the computer name.

Observed/expected result: DC01.

```
ipconfig /all
```

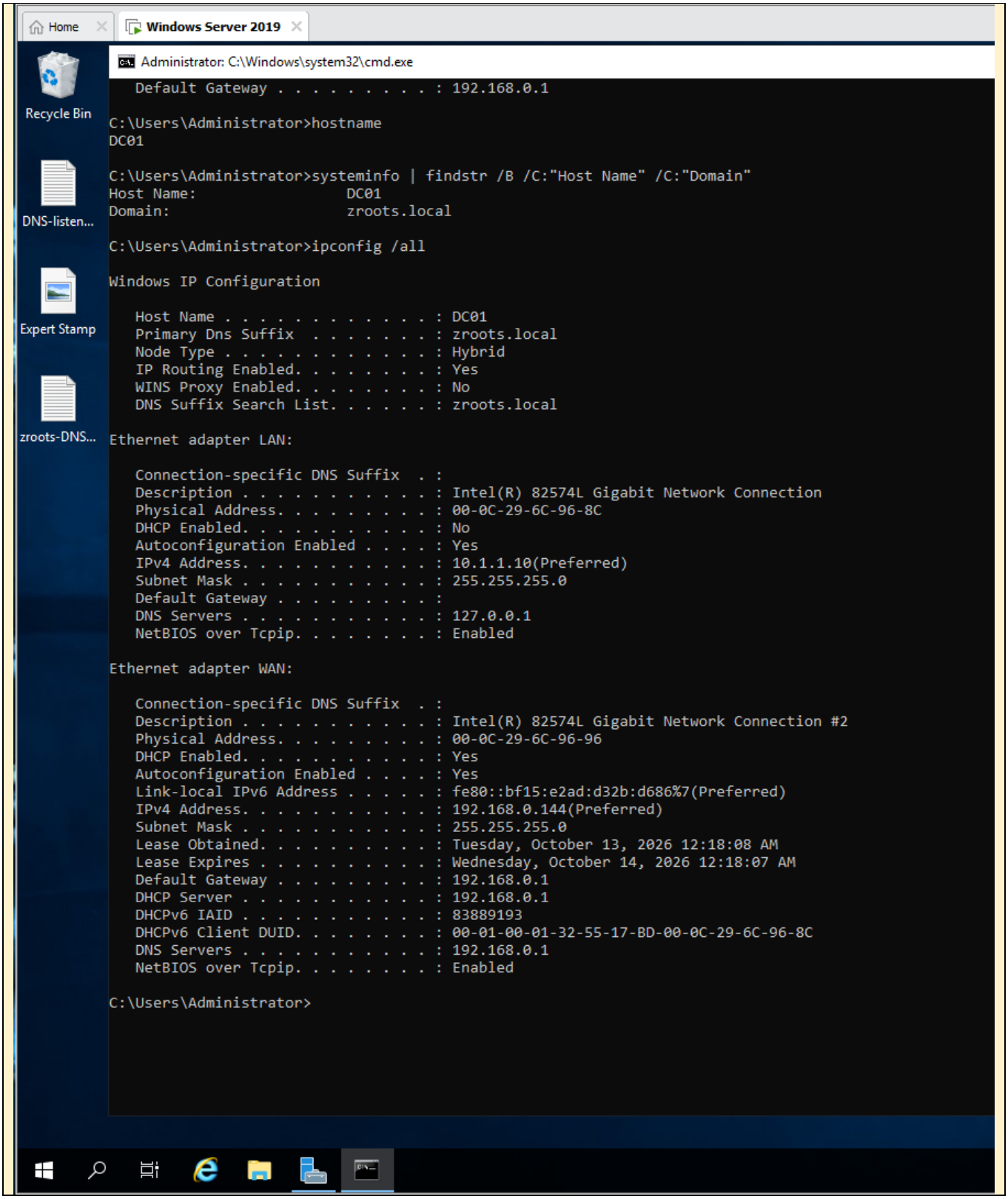
Purpose: Shows interface addresses, masks, gateways, DNS, DHCP state, MAC addresses and routing status.

Observed/expected result: LAN 10.1.1.10/24, DNS 127.0.0.1, no gateway; WAN 192.168.0.144/24, gateway/DNS 192.168.0.1; IP Routing Enabled: Yes.

```
systeminfo | findstr /B /C:"Domain"
```

Purpose: Filters systeminfo to the line beginning with Domain; /B anchors the match and /C specifies the exact string.

Observed/expected result: Domain: zroots.local.



7.2 Service status

```
sc query ntds
```

Purpose: Queries Service Control Manager for NTDS.

Observed/expected result: NTDS was RUNNING.

```
powershell -NoProfile -Command "Get-Service NTDS,DNS,Netlogon,KDC | Format-Table Name,Status,StartType -
AutoSize"
```

Purpose: Runs PowerShell without the profile, checks four services and formats their state/start type.

Observed/expected result: NTDS, DNS, Netlogon and KDC were Running and Automatic.

```
powershell -NoProfile -Command "Get-Service RemoteAccess,SharedAccess,DNS,NTDS"
```

Purpose: Checks RRAS, ICS, DNS and AD DS services.

Observed/expected result: DNS, NTDS and RemoteAccess Running/Automatic; SharedAccess Stopped/Disabled, consistent with RRAS rather than ICS.

```
powershell -NoProfile -Command "Get-Service RemoteAccess | Format-List Name,Status,StartType"
```

Purpose: Displays RRAS service details.

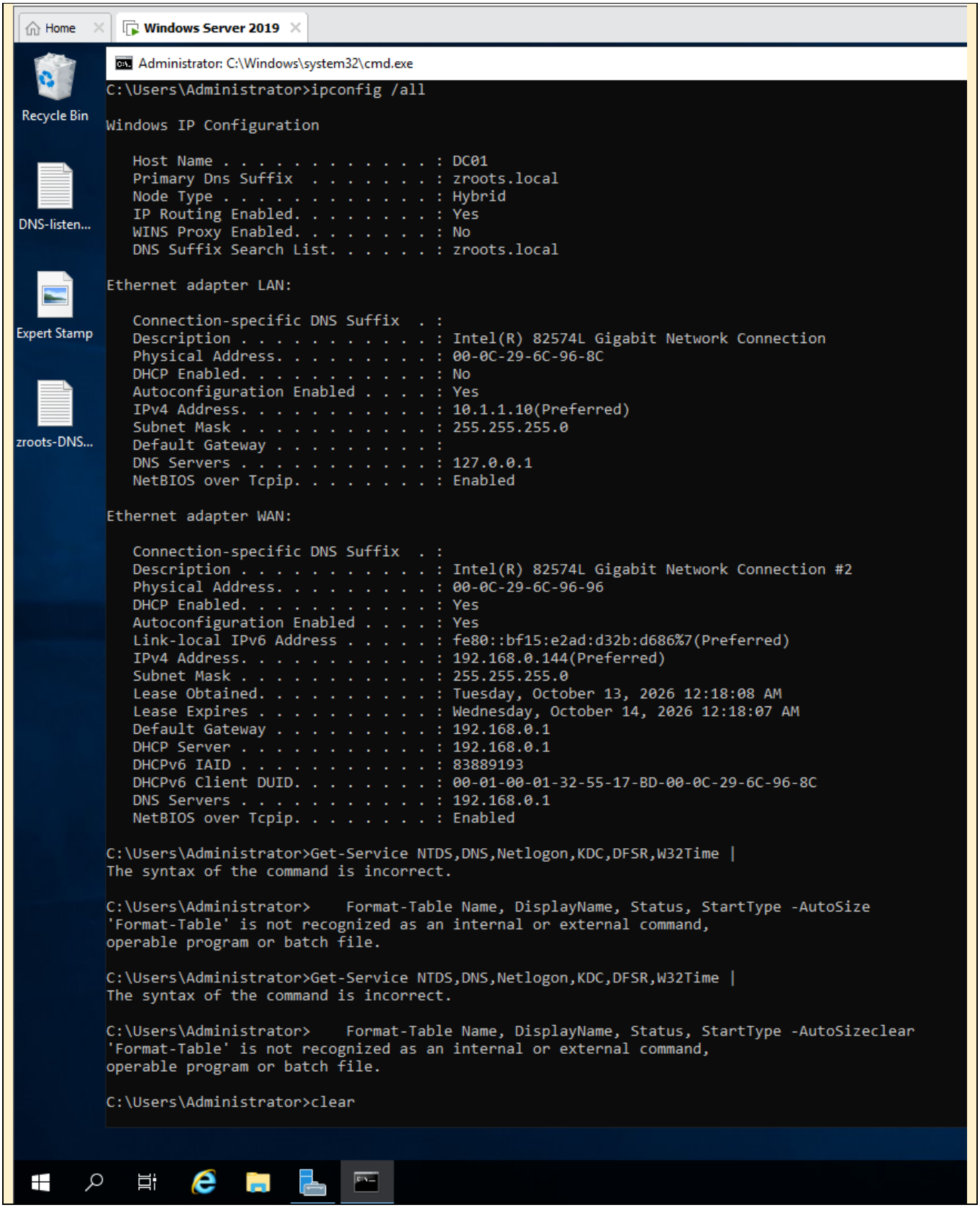
Observed/expected result: RemoteAccess Running, Automatic.

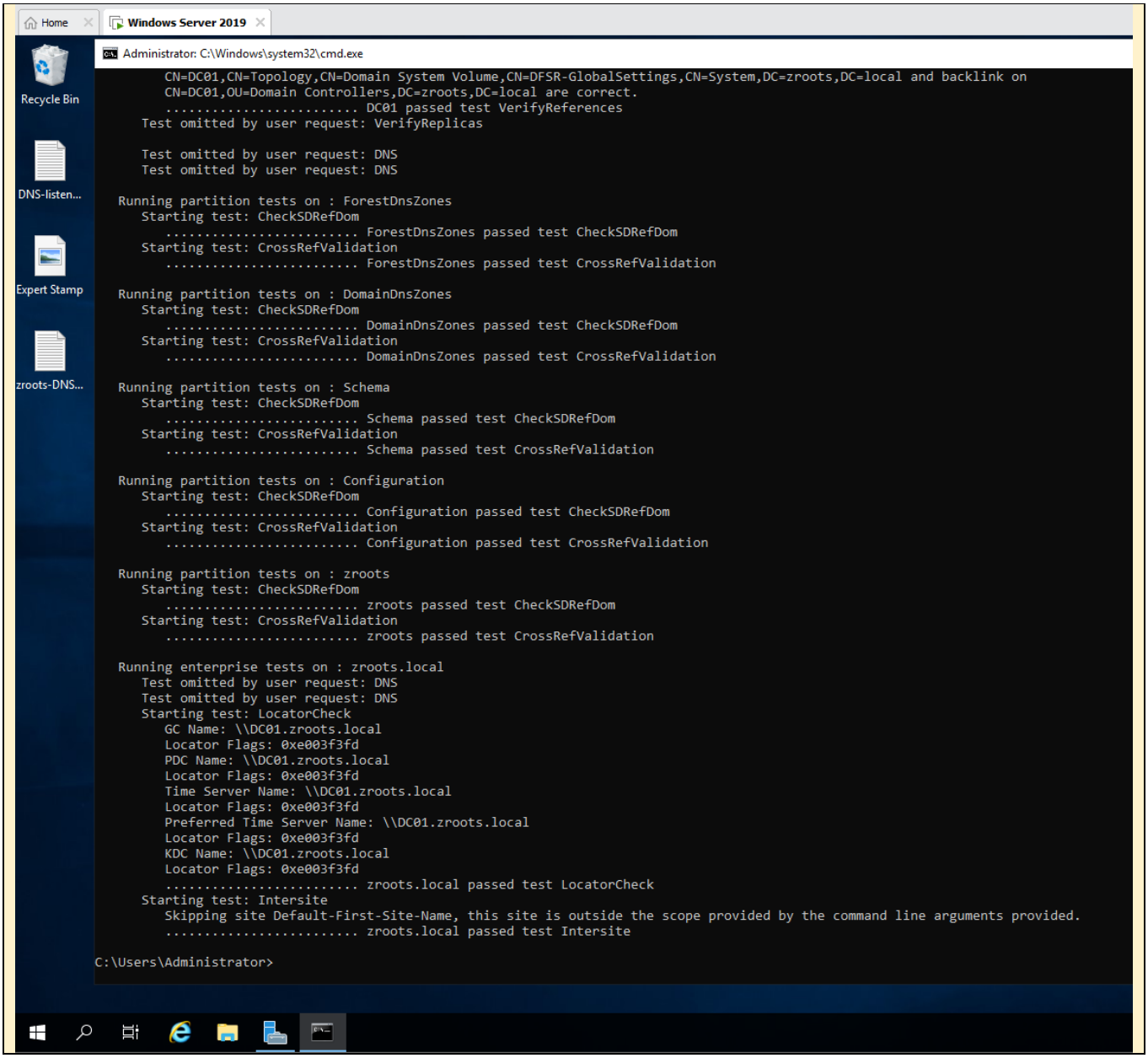
```
dcdiag /test:Advertising /test:Services /test:DNS
```

Purpose: Runs selected DC advertising, service and DNS tests.

Observed/expected result: Core Connectivity, Advertising, Services and DNS tests passed. Warnings were recorded for WAN adapter dynamic IP/upstream DNS and external reverse lookups.

DCDIAG warnings included the WAN adapter 192.168.0.144 being dynamic; 192.168.0.1 not being appropriate as the AD DNS test server; failure to query _ldap._tcp.zroots.local on that upstream resolver; and reverse lookup failures against public IPv6 root DNS servers. These warnings do not, by themselves, prove internal AD DNS is broken.


Home
Windows Server 2019
Recycle Bin
DNS-listen...
Expert Stamp
zroots-DNS...
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Administrator>ipconfig /all
Windows IP Configuration
Host Name . . . . . : DC01
Primary Dns Suffix . . . . . : zroots.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : Yes
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : zroots.local
Ethernet adapter LAN:
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-6C-96-8C
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 10.1.1.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DNS Servers . . . . . : 127.0.0.1
NetBIOS over Tcpip. . . . . : Enabled
Ethernet adapter WAN:
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) 82574L Gigabit Network Connection #2
Physical Address. . . . . : 00-0C-29-6C-96-96
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::bf15:e2ad:d32b:d686%7(Preferred)
IPv4 Address. . . . . : 192.168.0.144(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, October 13, 2026 12:18:08 AM
Lease Expires . . . . . : Wednesday, October 14, 2026 12:18:07 AM
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 83889193
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-55-17-BD-00-0C-29-6C-96-8C
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\Administrator>Get-Service NTDS,DNS,Netlogon,KDC,DFSR,W32Time |
The syntax of the command is incorrect.
C:\Users\Administrator> Format-Table Name, DisplayName, Status, StartType -AutoSize
'Format-Table' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Administrator>Get-Service NTDS,DNS,Netlogon,KDC,DFSR,W32Time |
The syntax of the command is incorrect.
C:\Users\Administrator> Format-Table Name, DisplayName, Status, StartType -AutoSizeclear
'Format-Table' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Administrator>clear



8. Domain Controller Discovery and LDAP SRV Verification

```
nltest /dsgetdc:zroot.local
```

Purpose: Initial discovery attempt with a misspelled domain (missing final s).

Observed/expected result: Failed with ERROR_NO_SUCH_DOMAIN / 1355; this was a spelling error.

```
nltest /dsgetdc:zroots.local
```

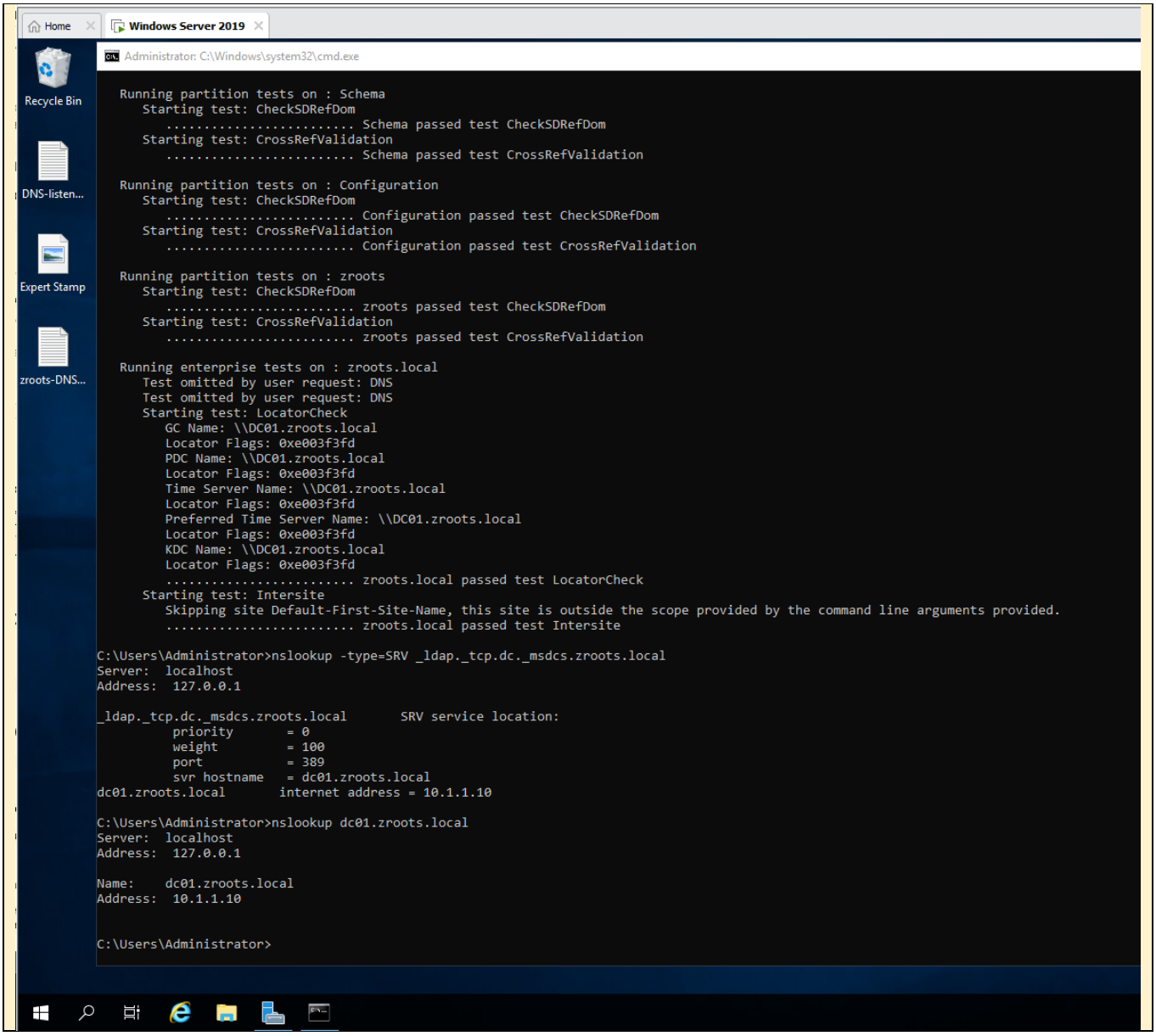
Purpose: Locates a domain controller for the exact domain.

Observed/expected result: Succeeded; \DC01.zroots.local at 10.1.1.10, domain/forest zroots.local.

```
nslookup -type=SRV _ldap._tcp.dc._msdcs.zroots.local 127.0.0.1
```

Purpose: Queries local DNS for the LDAP service-location record; -type=SRV requests service metadata.

Observed/expected result: Priority 0, weight 100, port 389, target dc01.zroots.local, target IP 10.1.1.10.



9. DNS Registration and Record Remediation

9.1 Initial findings

```
nslookup zroots.local
```

Purpose: Looks up the domain zone root.

Observed/expected result: Initially returned both 10.1.1.10 and 192.168.0.144.

```
nslookup DC01.zroots.local
```

Purpose: Looks up the domain controller FQDN.

Observed/expected result: DNS inventory initially contained A records for 10.1.1.10 and 192.168.0.144.

```
powershell -NoProfile -Command "Get-DnsServerResourceRecord -ZoneName 'zroots.local' -Name 'DC01' -RRType A"
```

Purpose: Lists DC01 A records in zroots.local.

Observed/expected result: Initially included LAN and WAN IPs.

```
powershell -NoProfile -Command "Get-DnsClient -InterfaceAlias 'LAN','WAN' | Format-Table
InterfaceAlias,RegisterThisConnectionsAddress,ConnectionSpecificSuffix -AutoSize"
```

Purpose: Inspects interface DNS registration.

Observed/expected result: Initially LAN and WAN both showed registration enabled.

9.2 Disable WAN registration and export inventory

```
powershell -NoProfile -Command "Set-DnsClient -InterfaceAlias 'WAN' -RegisterThisConnectionsAddress $false"
```

Purpose: Prevents the WAN interface from dynamically registering its address in DNS.

Observed/expected result: Later verified LAN=True, WAN=False.

```
powershell -NoProfile -Command "Get-DnsServerResourceRecord -ZoneName 'zroots.local' | Where-Object
{$_.RecordType -eq 'A'} | Select-Object
HostName,Timestamp,@{Name='IPv4Address';Expression={$_.RecordData.IPv4Address}} | Format-Table -AutoSize |
Out-File 'C:\Users\Administrator\Desktop\zroots-DNS-A-records-before.txt' -Width 160"
```

Purpose: Exports A records, timestamps and IPv4 addresses to a before-state text file.

Observed/expected result: Command completed; inventory saved to Administrator desktop.

9.3 Exact DNS record cleanup

The session removed the WAN address 192.168.0.144 from four exact record locations: @, DomainDnsZones and ForestDnsZones in zroots.local, and gc in _msdcs.zroots.local. Valid 10.1.1.10 records and AD SRV records were preserved.

```
powershell -NoProfile -Command "$r = Get-DnsServerResourceRecord -ZoneName 'zroots.local' -Name 'DC01' -RRType
A | Where-Object {$_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144'}; if ($r) { $r | Remove-
DnsServerResourceRecord -ZoneName 'zroots.local' -Force; 'Removed the DC01 WAN A record.' } else { 'No
matching DC01 WAN A record found.' }"
```

Purpose: Removes only DC01 A records matching the exact WAN IP.

Observed/expected result: Removed the DC01 WAN A record.

```
powershell -NoProfile -Command "$r = Get-DnsServerResourceRecord -ZoneName '_msdcs.zroots.local' -Name 'gc' -
RRType A | Where-Object {$_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144'}; if ($r) { $r |
Remove-DnsServerResourceRecord -ZoneName '_msdcs.zroots.local' -Force; 'Removed the WAN A record for gc.' }
else { 'No matching WAN A record found.' }"
```

Purpose: Removes only the Global Catalog gc record matching the WAN IP.

Observed/expected result: Removed the WAN A record for gc.

```
powershell -NoProfile -Command "$r = Get-DnsServerResourceRecord -ZoneName 'zroots.local' -Name '@' -RRType A
| Where-Object {$_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144'}; if ($r) { $r | Remove-
DnsServerResourceRecord -ZoneName 'zroots.local' -Force; 'Removed the WAN A record for the zone root.' } else
{ 'No matching WAN A record found.' }"
```

Purpose: Removes only the zone-root A record matching the WAN IP.

Observed/expected result: Removed the WAN A record for the zone root.

```
powershell -NoProfile -Command "$r = Get-DnsServerResourceRecord -ZoneName 'zroots.local' -Name
'DomainDnsZones' -RRType A | Where-Object {$_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144'};
if ($r) { $r | Remove-DnsServerResourceRecord -ZoneName 'zroots.local' -Force; 'Removed the DomainDnsZones WAN
A record.' } else { 'No matching DomainDnsZones WAN A record found.' }"
```

Purpose: Removes only the DomainDnsZones record matching the WAN IP.

Observed/expected result: Removed the DomainDnsZones WAN A record.

```
powershell -NoProfile -Command "$r = Get-DnsServerResourceRecord -ZoneName 'zroots.local' -Name
'ForestDnsZones' -RRType A | Where-Object {$_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144'};
if ($r) { $r | Remove-DnsServerResourceRecord -ZoneName 'zroots.local' -Force; 'Removed the ForestDnsZones WAN
A record.' } else { 'No matching ForestDnsZones WAN A record found.' }"
```

Purpose: Removes only the ForestDnsZones record matching the WAN IP.

Observed/expected result: Removed the ForestDnsZones WAN A record.

```
powershell -NoProfile -Command "$zones =
@(@{Zone='zroots.local';Name='@'},@{Zone='zroots.local';Name='DomainDnsZones'},@{Zone='zroots.local';Name='For
estDnsZones'},@{Zone='_msdcs.zroots.local';Name='gc'}); foreach ($z in $zones) { $records = Get-
DnsServerResourceRecord -ZoneName $z.Zone -Name $z.Name -RRType A -ErrorAction SilentlyContinue | Where-Object
{ $_.RecordData.IPv4Address.IPAddressToString -eq '192.168.0.144' }; foreach ($r in $records) { Remove-
DnsServerResourceRecord -ZoneName $z.Zone -InputObject $r -Force; Write-Output ('Removed WAN record: ' +
$z.Zone + ' / ' + $z.Name) } }"
```

Purpose: Final combined cleanup of the four exact zone/name combinations, filtered by WAN IP.

Observed/expected result: Output confirmed all four WAN records were removed. Some records had reappeared after earlier individual deletions; the combined cleanup was followed by successful verification.

9.4 Verify DNS records

```
nslookup DC01.zroots.local 127.0.0.1
```

Purpose: Queries DC01 against local DNS.

Observed/expected result: Returned only 10.1.1.10 after correction.

```
nslookup -type=A gc._msdcs.zroots.local 127.0.0.1
```

Purpose: Queries the GC A record against local DNS.

Observed/expected result: Returned only 10.1.1.10 after correction.

```
nslookup zroots.local 127.0.0.1
```

Purpose: Queries zone root against local DNS.

Observed/expected result: Returned only 10.1.1.10 after correction.

```
powershell -NoProfile -Command "Resolve-DnsName zroots.local -Server 127.0.0.1 -Type A; Resolve-DnsName
gc._msdcs.zroots.local -Server 127.0.0.1 -Type A"
```

Purpose: Queries both A records explicitly against local DNS.

Observed/expected result: Final output: both names resolved to 10.1.1.10 only.

```
powershell -NoProfile -Command "Get-DnsClient | Where-Object {$_.InterfaceAlias -in @('LAN','WAN')} | Format-
Table InterfaceAlias,RegisterThisConnectionsAddress,ConnectionSpecificSuffix -AutoSize"
```

Purpose: Confirms interface registration setting.

Observed/expected result: LAN True; WAN False.

9.5 Configure DNS listening address

```
powershell -NoProfile -Command "Set-DnsServerSetting -ListeningIpAddress 10.1.1.10"
```

Purpose: Attempted direct parameter form; unsupported on this server version.

Observed/expected result: Failed with parameter error; documented as a failed attempt, not final configuration.

```
powershell -NoProfile -Command "$s = Get-DnsServerSetting -All; $s.ListeningIPAddress = @('10.1.1.10'); Set-DnsServerSetting -InputObject $s"
```

Purpose: Reads the DNS settings object, sets ListeningIPAddress to the internal IP, and applies the object.

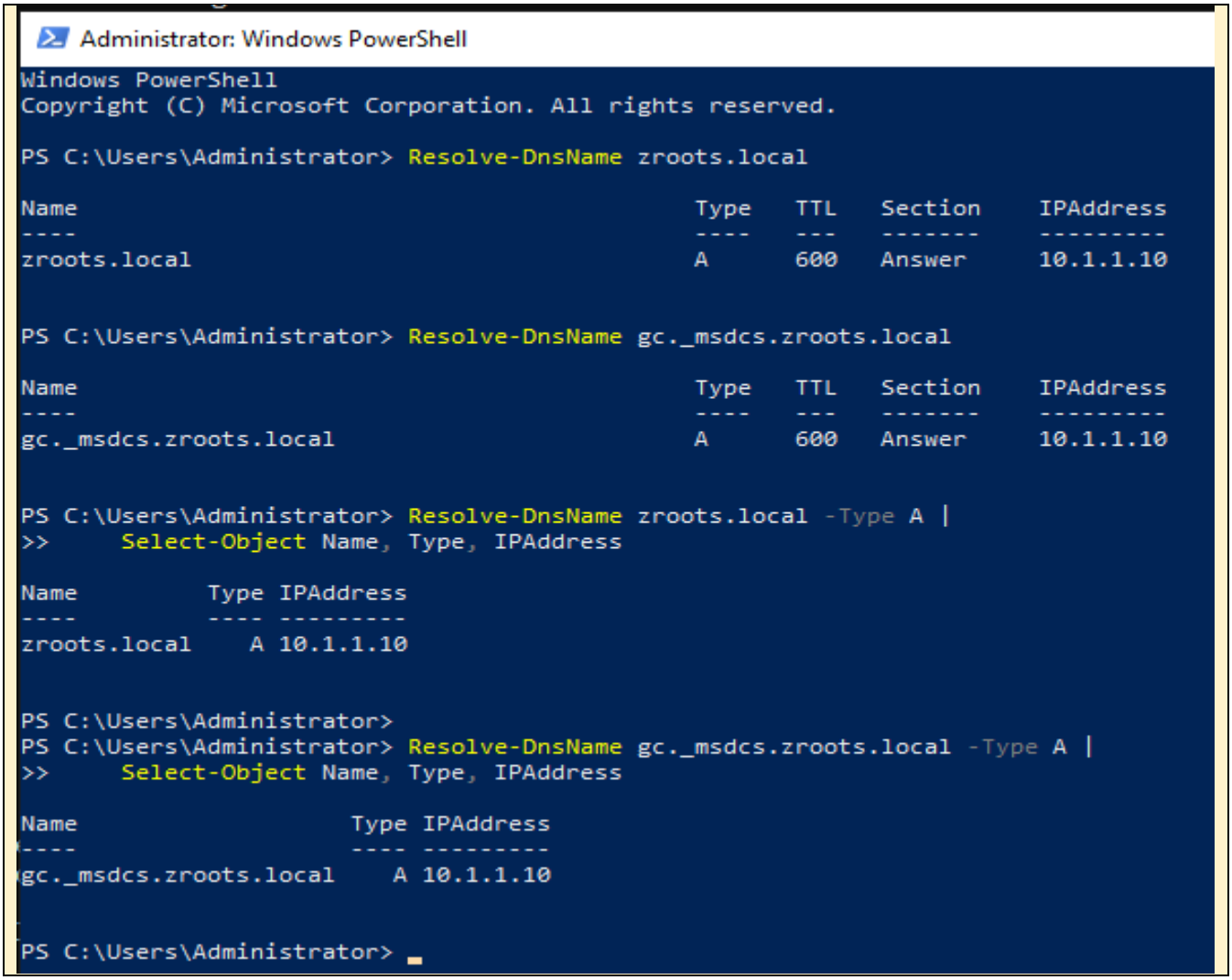
Observed/expected result: A -WhatIf preview succeeded before applying; final verification showed 10.1.1.10.

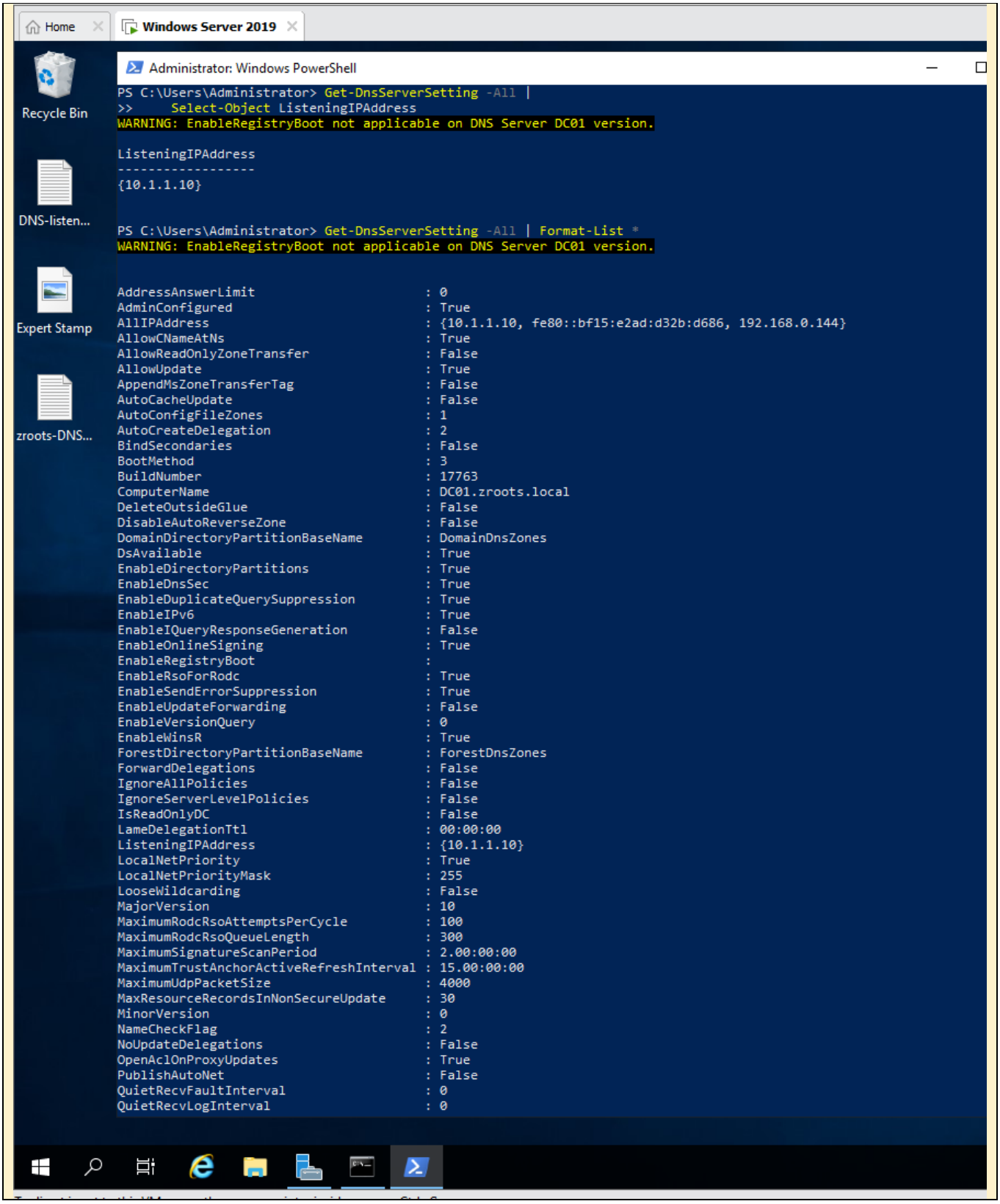
```
powershell -NoProfile -Command "Get-DnsServerSetting -All | Select-Object -ExpandProperty ListeningIPAddress"
```

Purpose: Displays the configured DNS listener addresses.

Observed/expected result: Output: 10.1.1.10.

AllIPAddress may still show all addresses assigned to the computer; ListeningIPAddress is the configured DNS listener list. WAN DNS client configuration remains 192.168.0.1 for upstream resolution.




Home
Windows Server 2019
Recycle Bin
DNS-listen...
Expert Stamp
zroots-DNS...
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-DnsServerSetting -All |
>> Select-Object ListeningIPAddress
WARNING: EnableRegistryBoot not applicable on DNS Server DC01 version.
ListeningIPAddress
{10.1.1.10}
PS C:\Users\Administrator> Get-DnsServerSetting -All | Format-List *
WARNING: EnableRegistryBoot not applicable on DNS Server DC01 version.
AddressAnswerLimit : 0
AdminConfigured : True
AllIPAddress : {10.1.1.10, fe80::bf15:e2ad:d32b:d686, 192.168.0.144}
AllowCNameAtNs : True
AllowReadOnlyZoneTransfer : False
AllowUpdate : True
AppendMsZoneTransferTag : False
AutoCacheUpdate : False
AutoConfigFileZones : 1
AutoCreateDelegation : 2
BindSecondaries : False
BootMethod : 3
BuildNumber : 17763
ComputerName : DC01.zroots.local
DeleteOutsideGlue : False
DisableAutoReverseZone : False
DomainDirectoryPartitionBaseName : DomainDnsZones
DsAvailable : True
EnableDirectoryPartitions : True
EnableDnsSec : True
EnableDuplicateQuerySuppression : True
EnableIPv6 : True
EnableIQueryResponseGeneration : False
EnableOnlineSigning : True
EnableRegistryBoot :
EnableRsoForRodc : True
EnableSendErrorSuppression : True
EnableUpdateForwarding : False
EnableVersionQuery : 0
EnableWinsR : True
ForestDirectoryPartitionBaseName : ForestDnsZones
ForwardDelegations : False
IgnoreAllPolicies : False
IgnoreServerLevelPolicies : False
IsReadOnlyDC : False
LameDelegationTtl : 00:00:00
ListeningIPAddress : {10.1.1.10}
LocalNetPriority : True
LocalNetPriorityMask : 255
LooseWildcarding : False
MajorVersion : 10
MaximumRodcRsoAttemptsPerCycle : 100
MaximumRodcRsoQueueLength : 300
MaximumSignatureScanPeriod : 2.00:00:00
MaximumTrustAnchorActiveRefreshInterval : 15.00:00:00
MaximumUdpPacketSize : 4000
MaxResourceRecordsInNonSecureUpdate : 30
MinorVersion : 0
NameCheckFlag : 2
NoUpdateDelegations : False
OpenAclOnProxyUpdates : True
PublishAutoNet : False
QuietRecvFaultInterval : 0
QuietRecvLogInterval : 0

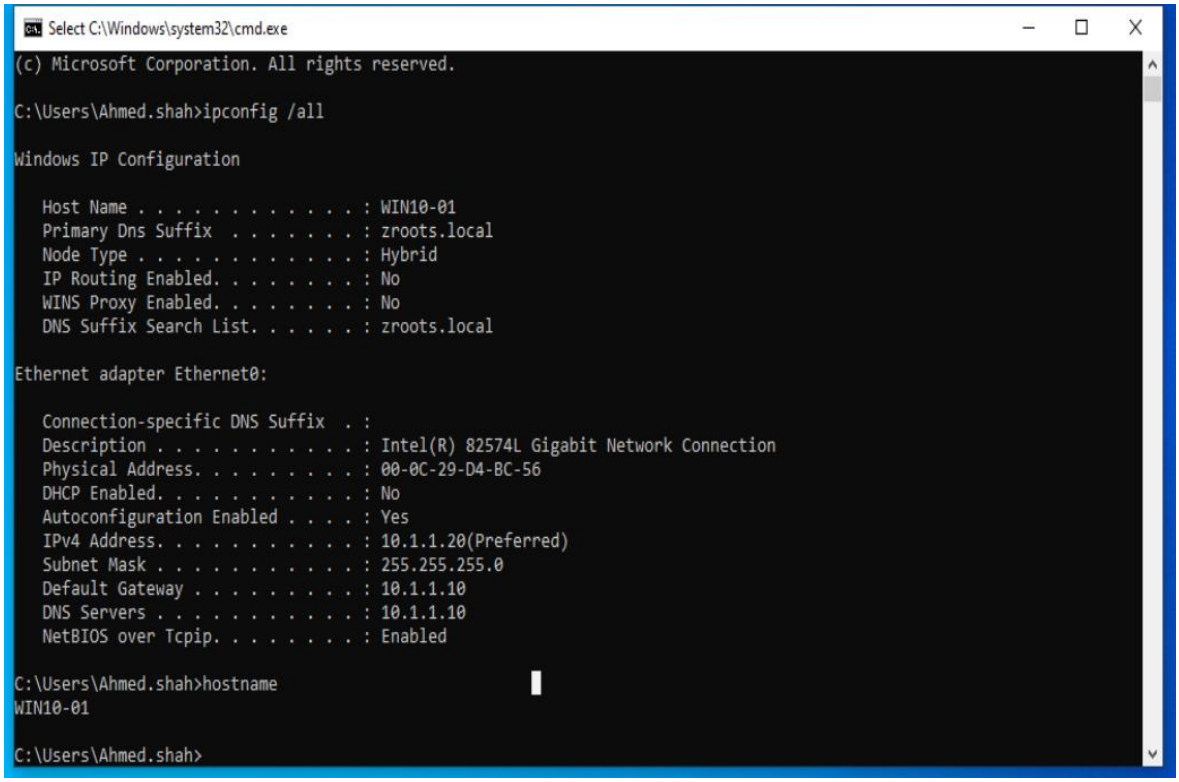


Figure 10.1 — WIN10-01 IP configuration and hostname

10. WIN10-01 Network Configuration

```
hostname
```

Purpose: Displays the client name.

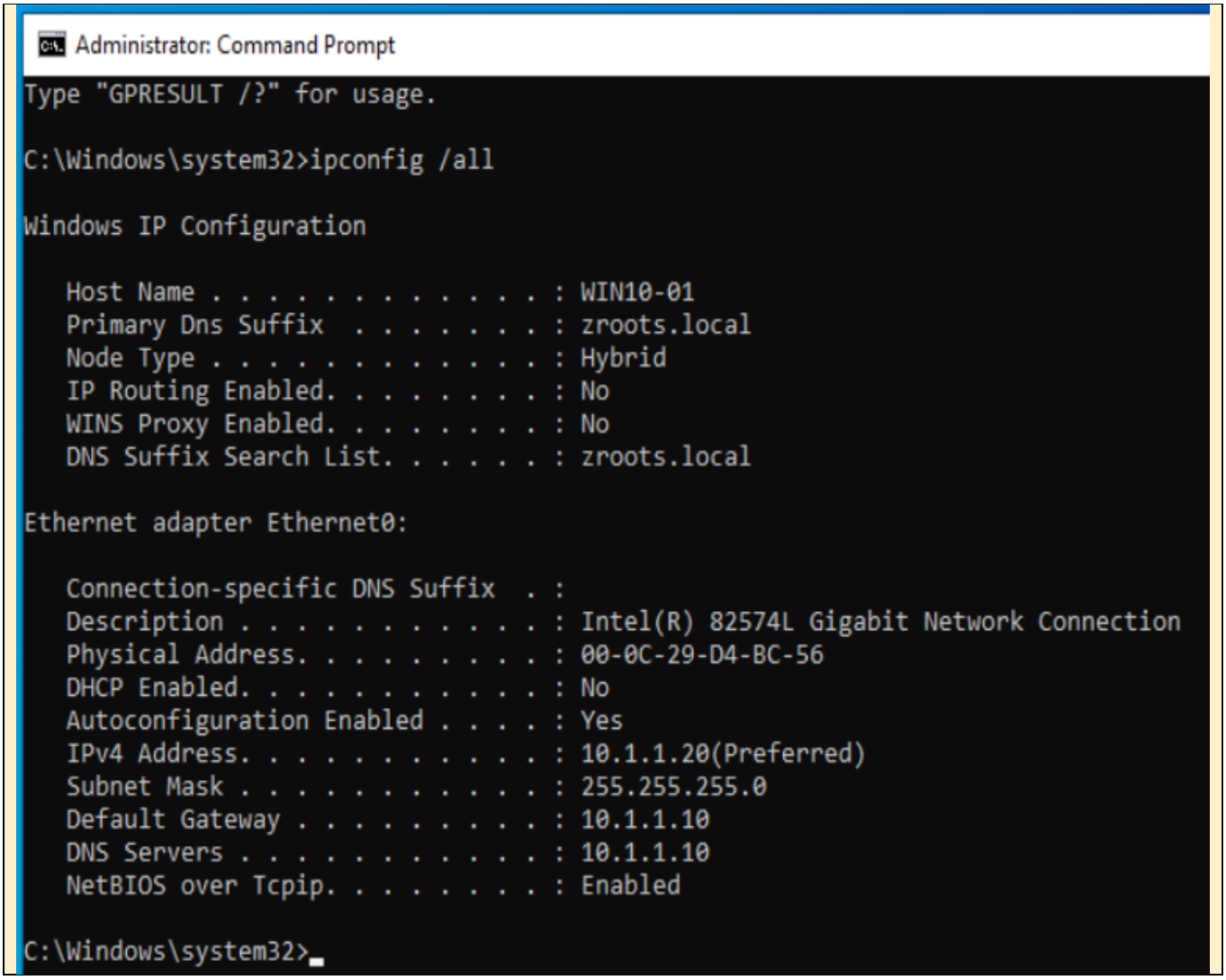
Observed/expected result: WIN10-01.

```
ipconfig /all
```

Purpose: Displays client network settings.

Observed/expected result: Observed: suffix zroots.local; Ethernet0 10.1.1.20 (Preferred); mask 255.255.255.0; gateway 10.1.1.10; DNS 10.1.1.10; DHCP disabled; DNS suffix search list zroots.local; NetBIOS over TCP/IP enabled; MAC 00-0C-29-D4-BC-56.

Setting	Observed value	Assessment
Computer	WIN10-01	Correct
Primary DNS suffix	zroots.local	Correct
IPv4	10.1.1.20	Correct
Mask	255.255.255.0	Correct
Gateway	10.1.1.10	Correct
DNS server	10.1.1.10	Correct
DHCP	Disabled	Matches static plan
DNS suffix search list	zroots.local	Correct
MAC	00-0C-29-D4-BC-56	Recorded



11. Client Connectivity and Domain Membership

```
ping 10.1.1.10
```

Purpose: Sends four default ICMP echo requests to DC01.

Observed/expected result: 4 sent, 4 received, 0% loss; replies under 1 ms; TTL 128.

```
nltest /dsgetdc:zroots.local
```

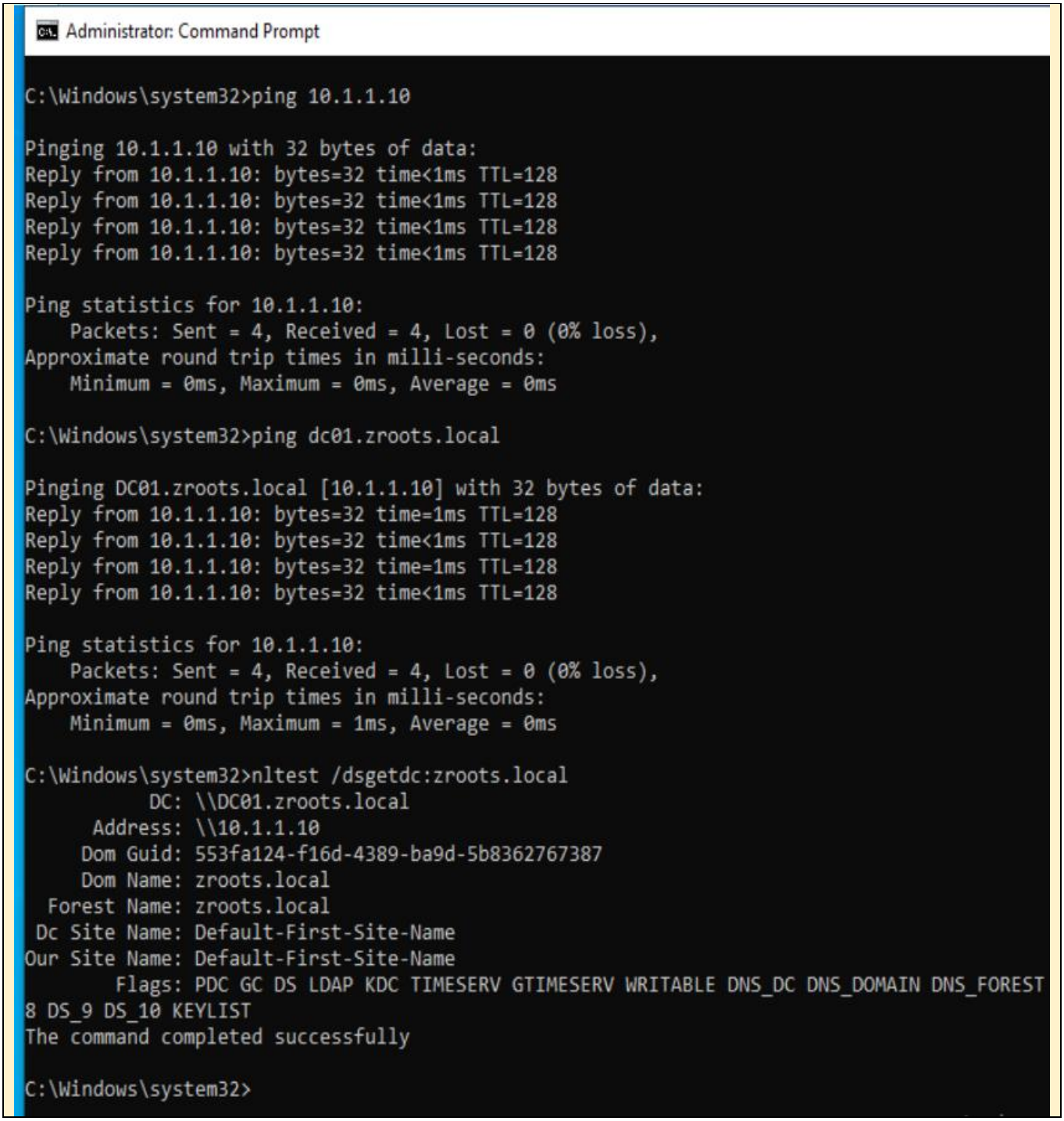
Purpose: Locates a domain controller for zroots.local.

Observed/expected result: DC \DC01.zroots.local at 10.1.1.10; command completed successfully.

```
systeminfo | findstr /B /C:"Domain"
```

Purpose: Displays the Domain line.

Observed/expected result: Domain: zroots.local.



12. Internet Connectivity and External DNS

```
ping 192.168.0.1
```

Purpose: Tests reachability to the physical router from the client.

Observed/expected result: 4/4 replies; 0% loss; average 1 ms.

```
ping 1.1.1.1
```

Purpose: Tests external IPv4 connectivity without needing DNS.

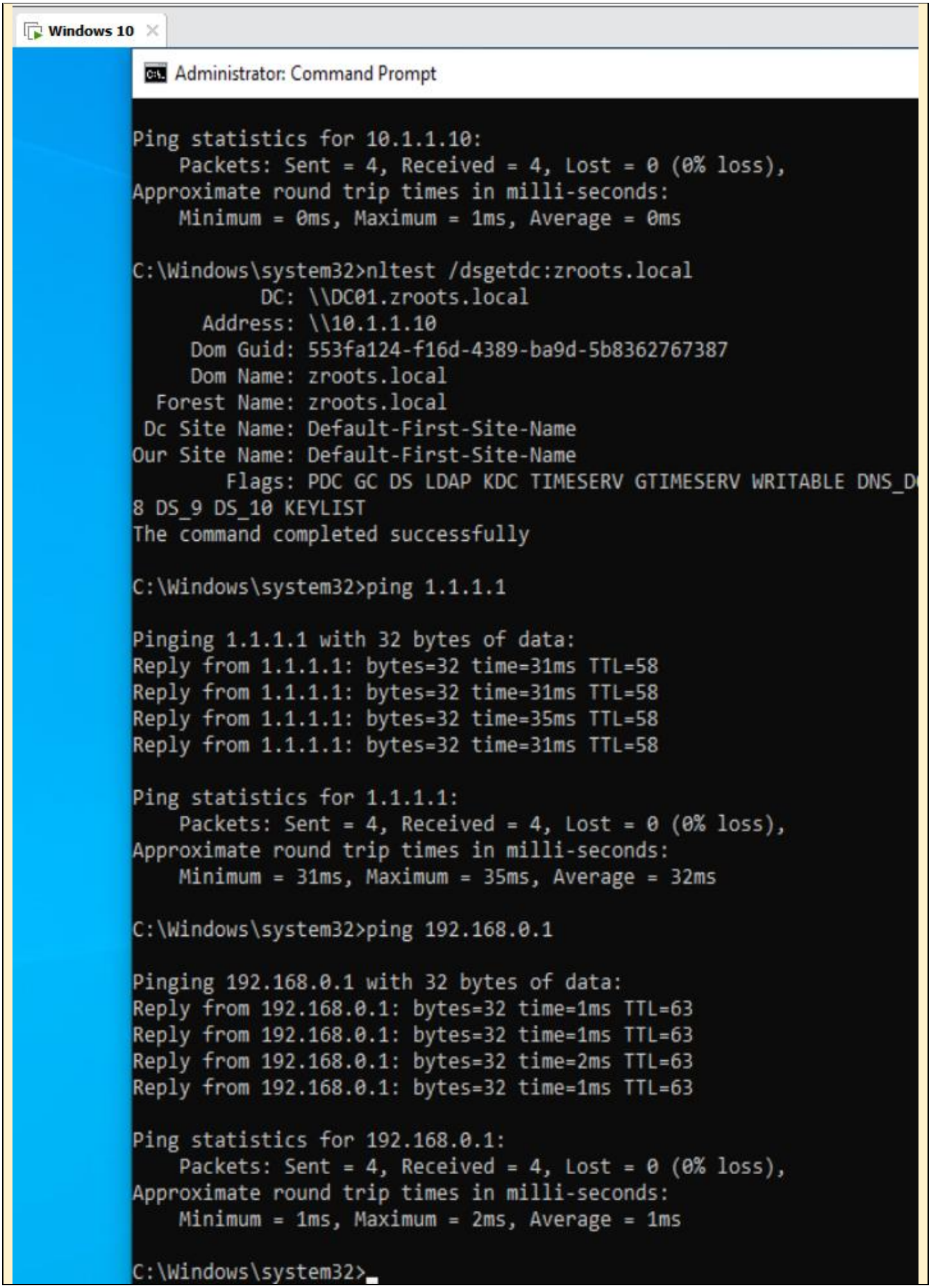
Observed/expected result: 4/4 replies; 0% loss; min 33 ms, max 35 ms, average 34 ms.

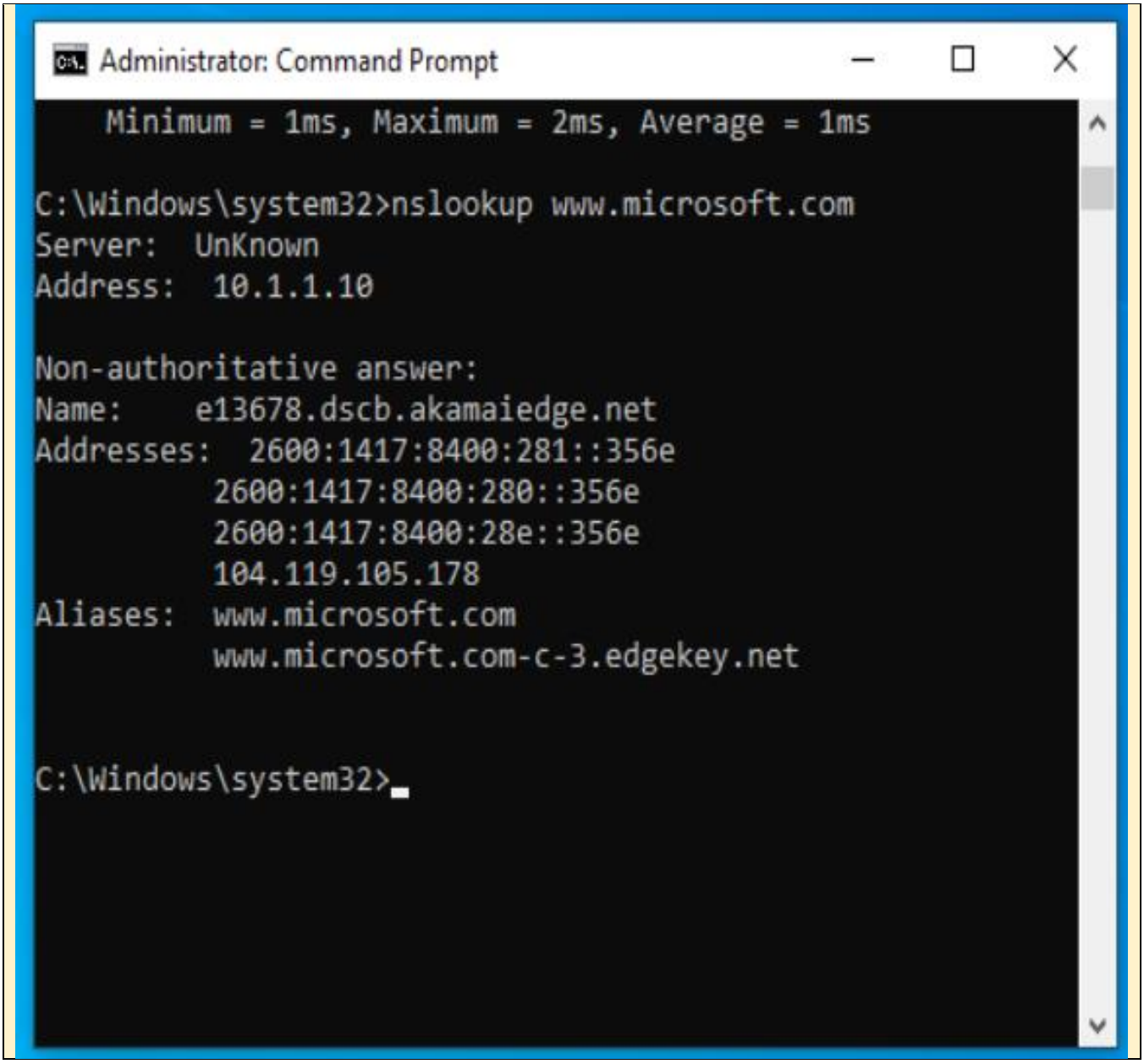
```
nslookup www.microsoft.com
```

Purpose: Queries the configured DNS server for an external hostname.

Observed/expected result: Server address 10.1.1.10; resolved through aliases www.microsoft.com-c-3.edgekey.net and e13678.dscb.akamaiedge.net; IPv4 23.41.38.100 and several IPv6 addresses.

The nslookup label “Server: UnKnown” did not indicate failure: the DNS server at 10.1.1.10 returned a valid non-authoritative answer. Non-authoritative is normal for recursive lookups when the resolver is not authoritative for Microsoft’s zone.


Windows 10
Administrator: Command Prompt
Ping statistics for 10.1.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Windows\system32>nltest /dsgetdc:zroots.local
DC: \\DC01.zroots.local
Address: \\10.1.1.10
Dom Guid: 553fa124-f16d-4389-ba9d-5b8362767387
Dom Name: zroots.local
Forest Name: zroots.local
Dc Site Name: Default-First-Site-Name
Our Site Name: Default-First-Site-Name
Flags: PDC GC DS LDAP KDC TIMESERV GTIMESERV WRITABLE DNS_D
8 DS_9 DS_10 KEYLIST
The command completed successfully
C:\Windows\system32>ping 1.1.1.1
Pinging 1.1.1.1 with 32 bytes of data:
Reply from 1.1.1.1: bytes=32 time=31ms TTL=58
Reply from 1.1.1.1: bytes=32 time=31ms TTL=58
Reply from 1.1.1.1: bytes=32 time=35ms TTL=58
Reply from 1.1.1.1: bytes=32 time=31ms TTL=58
Ping statistics for 1.1.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 31ms, Maximum = 35ms, Average = 32ms
C:\Windows\system32>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=63
Reply from 192.168.0.1: bytes=32 time=1ms TTL=63
Reply from 192.168.0.1: bytes=32 time=2ms TTL=63
Reply from 192.168.0.1: bytes=32 time=1ms TTL=63
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Windows\system32>



13. Consolidated Results and Evidence Register

Test	Observed result	Status
DC01 identity/domain	DC01; zroots.local	PASS
Core services	NTDS, DNS, Netlogon, KDC running/Automatic	PASS
RRAS	RemoteAccess running/Automatic	PASS
DCDIAG selected tests	Core tests passed; warnings documented	PASS WITH WARNINGS
Domain controller discovery	DC01.zroots.local at 10.1.1.10	PASS
LDAP SRV	dc01.zroots.local port 389 → 10.1.1.10	PASS
WAN DNS record cleanup	Final direct queries returned internal IP only	PASS
DNS listener	10.1.1.10	PASS

Client static configuration	10.1.1.20/24; gateway/DNS 10.1.1.10	PASS
Client to DC01	4/4 replies; 0% loss	PASS
Domain discovery/membership	DC01 discovered; zroots.local reported	PASS
Router reachability	4/4 replies; 0% loss	PASS
External IP	1.1.1.1 4/4 replies; 0% loss	PASS
External DNS	www.microsoft.com resolved through 10.1.1.10	PASS

Evidence should include the topology diagram, DC01 identity/service/DCDIAG output, before-state DNS inventory, DNS corrections and verification, listener configuration, and all WIN10-01 screenshots.

14. Troubleshooting and Edge Cases

Issue	Guidance
NLTEST error 1355	The first command misspelled zroots.local as zroot.local. Re-run with the exact domain spelling.
Domain name returned WAN and LAN IPs	Inspect authoritative A records and per-interface registration. Disable WAN registration and remove only exact, verified WAN A records.
WAN records reappeared	AD-integrated dynamic updates or interface registration may recreate records. Verify WAN registration is False, check the configured listener, clean only exact records and query again. Preserve SRV records.
Set-DnsServerSetting parameter error	The direct parameter form was unsupported. Use Get-DnsServerSetting -All, modify ListeningIPAddress and apply the settings object.
nslookup server is “UnKnown”	A friendly/reverse name may be unavailable; judge success by the returned answer and configured server address.
DCDIAG external/reverse lookup warnings	Check internal AD SRV and host records separately. External reverse lookup warnings do not alone prove internal DNS failure.
Ping fails but applications work	ICMP may be filtered; test the protocol/application in question.
Client cannot find domain	Verify IP, mask, gateway, DNS pointing to DC01, time, exact domain spelling, services and SRV records. Do not configure public DNS directly on the domain client.
External IP works but names fail	Check client DNS, use nslookup, and verify DC01 can resolve external names via upstream DNS/root hints.
dcdiag-test-record A record	A record 1.1.1.1 was observed during diagnostics. Verify whether it remains and its purpose before removing; it was not one of the four WAN records cleaned.

15. Security Lessons and Recommendations

- Prevent the WAN interface from registering its address into internal AD DNS when clients should use the internal LAN address.
- Domain clients should use AD DNS so domain-specific SRV and host records resolve correctly.
- Dual-homed domain controllers need careful DNS registration, listener and firewall scope.
- Keep Windows Defender Firewall enabled; allow only necessary services from trusted networks.
- Keep ICS disabled when RRAS is the selected routing design.
- Use exact DNS record filters; preserve SRV records and valid LAN A records.
- Capture before/after evidence for administrative changes.
- Never expose Metasploitable or other vulnerable targets to bridged WAN/public networks.
- Successful ping and DNS tests prove only the tested paths, not all ports, protocols or endpoints.
- Review WAN DHCP lease and DNS configuration periodically.

16. Knowledge Check

1. What does an LDAP SRV record do?

Answer: It advertises the host and port for LDAP so clients can locate directory services.

2. Why should DC01 resolve internally to 10.1.1.10?

Answer: That is the internal LAN address used by MyLabNet clients.

3. Why disable WAN DNS registration?

Answer: To prevent the WAN address from being dynamically published as an internal AD host address.

4. Bridged versus RRAS/NAT?

Answer: Bridged attaches a VM NIC to the physical LAN; RRAS/NAT routes/translates traffic between private LAN and WAN.

5. Does "Server: UnKnown" mean DNS failed?

Answer: No. The server returned a valid answer in this session.

6. What does ping 1.1.1.1 with 0% loss prove?

Answer: That the target answered the four ICMP probes; it does not prove every protocol works.

7. Why not use public DNS directly on an AD client?

Answer: It may bypass AD DNS records required for domain discovery and services.

8. Why remove records using exact filters?

Answer: To avoid deleting valid internal addresses or AD service records.

17. Lab Report and Conclusion

The baseline objective was achieved for the components tested. DC01's core directory/DNS services were running; selected DCDIAG tests passed with warnings; domain-controller discovery succeeded; and the LDAP SRV record targeted dc01.zroots.local at 10.1.1.10 on port 389. Incorrect WAN A records were removed, WAN DNS registration was disabled, and DNS listening was configured for 10.1.1.10. Final direct DNS queries returned the intended internal address.

WIN10-01 matched the documented static network settings, reached DC01 with zero packet loss, discovered the domain controller, reported membership in zroots.local, reached the physical router and external IP 1.1.1.1, and resolved www.microsoft.com through 10.1.1.10. No server or client restart was required. The next lab should preserve this baseline and avoid network changes unless a specific test justifies them.

Final assessment
LAB 01 PASSED for the tests documented here, with DCDIAG warnings retained for future review. This is not a claim that every endpoint or security control has been exhaustively audited.

Appendix A — Command Reference

Command	Host	Purpose
hostname	DC01 / WIN10-01	Display computer name.
ipconfig /all	Both	Full network configuration.
systeminfo | findstr /B /C:"Domain"	Both	Show domain line.
sc query ntds	DC01	Query AD DS service.
Get-Service NTDS,DNS,Netlogon,KDC (via PowerShell)	DC01	Check core services.
Get-Service RemoteAccess,SharedAccess,DNS,NTDS (via PowerShell)	DC01	Check RRAS/ICS/DNS/AD DS.
dcdiag /test:Advertising /test:Services /test:DNS	DC01	Selected DC diagnostics.
nltest /dsgetdc:zroot.local	DC01	Failed typo attempt.
nltest /dsgetdc:zroots.local	Both	Discover domain controller.
nslookup -type=SRV _ldap._tcp.dc._msdcs.zroots.local 127.0.0.1	DC01	LDAP SRV lookup.
nslookup zroots.local	DC01	Initial zone-root lookup.
nslookup DC01.zroots.local	DC01	DC A-record lookup.
Set-DnsClient -InterfaceAlias WAN -RegisterThisConnectionsAddress $false	DC01	Disable WAN DNS registration.
Get-DnsServerResourceRecord ... | Where-Object RecordType -eq A ... Out-File ...	DC01	Export A-record inventory.
Resolve-DnsName zroots.local -Server 127.0.0.1 -Type A	DC01	Verify zone-root A record.
Resolve-DnsName gc._msdcs.zroots.local -Server 127.0.0.1 -Type A	DC01	Verify GC A record.
Get-DnsServerSetting -All; set ListeningIPAddress; Set-DnsServerSetting -InputObject $s	DC01	Configure DNS listener.
ping 10.1.1.10	WIN10-01	Test DC connectivity.
nltest /dsgetdc:zroots.local	WIN10-01	Client DC discovery.
ping 192.168.0.1	WIN10-01	Router reachability.
ping 1.1.1.1	WIN10-01	External IP connectivity.
nslookup www.microsoft.com	WIN10-01	External DNS resolution.

Appendix B — Screenshot Checklist

- Topology diagram with WAN/LAN addresses.
- DC01 hostname and ipconfig /all.
- Service status for NTDS, DNS, Netlogon, KDC, RemoteAccess and SharedAccess.
- Completed DCDIAG output with warnings.
- Successful NLTEST on DC01.
- LDAP SRV output.
- Before-state DNS A-record inventory.
- WAN DNS registration setting: LAN True, WAN False.
- Combined cleanup output showing four WAN records removed.
- Final Resolve-DnsName output.
- DNS ListeningIPAddress output.
- WIN10-01 ipconfig /all.
- WIN10-01 ping/NLTEST/domain output.
- Router and external IP pings.
- External DNS lookup output.

Formatting and Submission Notes

- Fill the cover-page identity fields.
- Replace screenshot placeholders with original screenshots and captions.
- Keep the before-state DNS inventory with this report as supporting evidence.
- If exporting to PDF, inspect page breaks, tables, code blocks and screenshot sizing.